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Attenuator using variable resistors

Abstract

Certain aspects of the present disclosure are directed towards a signal processing circuit. The signal processing circuit generally includes: an impedance transformation circuit coupled between a first node and a second node; and an attenuation circuit including a first variable resistive element coupled between the first node and a reference potential node, and a second variable resistive element coupled between the second node and the reference potential node.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) Certain aspects of the present disclosure generally relate to electronic components and, more particularly, to circuitry for signal amplification and attenuation.

BACKGROUND

(2) Electronic devices include computing devices such as desktop computers, notebook computers, tablet computers, smartphones, wearable devices like a smartwatch, internet servers, and so forth. These various electronic devices provide information, entertainment, social interaction, security, safety, productivity, transportation, manufacturing, and other services to human users. These various electronic devices depend on wireless communications for many of their functions. Wireless communication systems and devices are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These

systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such systems include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, and orthogonal frequency division multiple access (OFDMA) systems (e.g., a Long Term Evolution (LTE) system or a New Radio (NR) system). Wireless devices may include transmitters for processing signals for transmission via antennas. A transmitter may include a power amplifier (PA) for amplifying a signal for transmission.

SUMMARY

(3) The systems, methods, and devices of the disclosure each have several aspects, no single one of which is solely responsible for its desirable attributes. Without limiting the scope of this disclosure as expressed by the claims which follow, some features will now be discussed briefly. After considering this discussion, and particularly after reading the section entitled “Detailed Description,” one will understand how the features of this disclosure provide the advantages described herein.

(4) Certain aspects of the present disclosure are directed towards a signal processing circuit. The signal processing circuit generally includes: an impedance transformation circuit coupled between a first node and a second node; and an attenuation circuit including a first variable resistive element coupled between the first node and a reference potential node, and a second variable resistive element coupled between the second node and the reference potential node.

(5) Certain aspects of the present disclosure are directed towards a wireless device. The wireless device generally includes: a controller configured to determine a power mode associated with signal transmission; an attenuator comprising a first variable resistive element coupled between a first node and a reference potential node, and a second variable resistive element coupled between a second node and the reference potential node, wherein an impedance transformation circuit is coupled between the first node and the second node, and wherein the controller is configured to bias the first variable resistive element and the second variable resistive element to generate an attenuator signal based on an input signal; and an amplifier coupled to the attenuator and configured to generate a signal for transmission based on the attenuator signal.

(6) Certain aspects of the present disclosure are directed towards a method for signal processing. The method generally includes receiving an input signal and generating, via an attenuation circuit, an attenuator signal based on the input signal, the attenuation circuit comprising: a first variable resistive element coupled between a first node and a reference potential node, and a second variable resistive element coupled between a second node and the reference potential node, wherein an impedance transformation circuit is coupled between the first node and the second node.

(7) To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the appended drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above-recited features of the present disclosure can be understood in detail, a more particular description, briefly summarized above, may be by reference to aspects, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only certain aspects of this disclosure and are therefore not to be considered limiting of its scope, for the description may admit to other equally effective aspects.

- (2) FIG. 1 is a diagram of an example wireless communications network, in accordance with certain aspects of the present disclosure.
- (3) FIG. 2 is a block diagram of an example access point (AP) and example user terminals, in accordance with certain aspects of the present disclosure.
- (4) FIG. 3 is a block diagram of an example transceiver front end, in accordance with certain aspects of the present disclosure.
- (5) FIG. 4 illustrates an example signal processing circuit, in accordance with certain aspects of the present disclosure.
- (6) FIG. 5 illustrates example impedance transformation circuits, in accordance with certain aspects of the present disclosure.
- (7) FIG. 6 illustrates an example amplification circuit including attenuation and impedance transformation circuits, in accordance with certain aspects of the present disclosure.
- (8) FIG. 7 is a flow diagram depicting example operations for signal processing, in accordance with certain aspects of the present disclosure.
- (9) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one aspect may be beneficially utilized on other aspects without specific recitation.

DETAILED DESCRIPTION

(10) Certain aspects of the present disclosure generally relate to a signal processing circuit, including an attenuation circuit. The attenuation circuit may include a first variable resistive element coupled between a first node and a reference potential node and a second variable resistive element coupled between a second node and the reference potential node. An impedance transformation circuit may be coupled between the first node and the second node. The signal processing circuit may be part of a transmitter that may operate in either a high-power mode (HPM) or a low-power mode (LPM). During the HPM, the first and second variable resistive elements may be configured with high impedance (e.g., effectively an open circuit), reducing any impact of the attenuation circuit on the HPM performance. In the LPM, the first and second variable resistive elements may be biased to generate an attenuated signal, which may be amplified by one or more power amplifiers (PAs) in the transmitter.

(11) Example Wireless Communications

(12) FIG. 1 illustrates a wireless communications system **100** with access points **110** and user terminals **120**, in which aspects of the present disclosure may be practiced. For simplicity, only one access point **110** is shown in FIG. 1. An access point (AP) is generally a fixed station that communicates with the user terminals and may also be referred to as a base station (BS), an evolved Node B (eNB), a next generation Node B (gNB), or some other terminology. A user terminal (UT) may be fixed or mobile and may also be referred to as a mobile station (MS), an access terminal, user equipment (UE), a station (STA), a client, a wireless device, or some other terminology. A user terminal may be a wireless device, such as a cellular phone, a personal digital assistant (PDA), a handheld device, a wireless modem, a laptop computer, a tablet, a personal computer, etc.

(13) Access point **110** may communicate with one or more user terminals **120** at any given moment on the downlink and uplink. The downlink (i.e., forward link) is the communication link from the access point to the user terminals, and the uplink (i.e., reverse link) is the communication link from the user terminals to the access point. A user terminal may also communicate peer-to-peer with another user terminal. A system controller **130** couples to and provides coordination and control for the access points.

(14) Wireless communications system **100** employs multiple transmit and multiple receive antennas for data transmission on the downlink and uplink. Access point **110** may be equipped with a number $N_{\text{sub.ap}}$ of antennas to achieve transmit diversity for downlink transmissions and/or receive diversity for uplink transmissions. A set N_u of selected user terminals **120** may receive

downlink transmissions and transmit uplink transmissions. Each selected user terminal transmits user-specific data to and/or receives user-specific data from the access point. In general, each selected user terminal may be equipped with one or multiple antennas (i.e., $N_{\text{sub.ut}} \geq 1$). The N_u selected user terminals can have the same or different number of antennas.

(15) Wireless communications system **100** may be a time division duplex (TDD) system or a frequency division duplex (FDD) system. For a TDD system, the downlink and uplink share the same frequency band. For an FDD system, the downlink and uplink use different frequency bands. Wireless communications system **100** may also utilize a single carrier or multiple carriers for transmission. Each user terminal **120** may be equipped with a single antenna (e.g., to keep costs down) or multiple antennas (e.g., where the additional cost can be supported).

(16) In some aspects, the user terminal **120** or access point **110** may include an attenuator having one or more variable resistive elements, which may be coupled to an input of an amplifier, as described in more detail herein.

(17) FIG. 2 shows a block diagram of access point **110** and two user terminals **120_m** and **120_x** in the wireless communications system **100**. Access point **110** is equipped with $N_{\text{sub.ap}}$ antennas **224_a** through **224_{ap}**. User terminal **120_m** is equipped with $N_{\text{sub.ut,m}}$ antennas **252_{ma}** through **252_{mu}**, and user terminal **120_x** is equipped with $N_{\text{sub.ut,x}}$ antennas **252_{xa}** through **252_{xu}**. Access point **110** is a transmitting entity for the downlink and a receiving entity for the uplink. Each user terminal **120** is a transmitting entity for the uplink and a receiving entity for the downlink. As used herein, a “transmitting entity” is an independently operated apparatus or device capable of transmitting data via a frequency channel, and a “receiving entity” is an independently operated apparatus or device capable of receiving data via a frequency channel. In the following description, the subscript “dn” denotes the downlink, the subscript “up” denotes the uplink, $N_{\text{sub.up}}$ user terminals are selected for simultaneous transmission on the uplink, $N_{\text{sub.dn}}$ user terminals are selected for simultaneous transmission on the downlink, $N_{\text{sub.up}}$ may or may not be equal to $N_{\text{sub.dn}}$, and $N_{\text{sub.up}}$ and $N_{\text{sub.dn}}$ may be static values or can change for each scheduling interval. Beam-steering, beamforming, or some other spatial processing technique may be used at the access point and/or user terminal.

(18) On the uplink, at each user terminal **120** selected for uplink transmission, a TX data processor **288** receives traffic data from a data source **286** and control data from a controller **280**. TX data processor **288** processes (e.g., encodes, interleaves, and modulates) the traffic data $\{d_{\text{sub.up}}\}$ for the user terminal based on the coding and modulation schemes associated with the rate selected for the user terminal and provides a data symbol stream $\{s_{\text{sub.up}}\}$ for one of the $N_{\text{sub.ut,m}}$ antennas. A transceiver front end (TX/RX) **254** (also known as a radio frequency front end (RFFE)) receives and processes (e.g., converts to analog, amplifies, filters, and frequency upconverts) a respective symbol stream to generate an uplink signal. The transceiver front end **254** may also route the uplink signal to one of the $N_{\text{sub.ut,m}}$ antennas for transmit diversity via an RF switch, for example. The controller **280** may control the routing within the transceiver front end **254**. Memory **282** may store data and program codes for the user terminal **120** and may interface with the controller **280**.

(19) A number $N_{\text{sub.up}}$ of user terminals **120** may be scheduled for simultaneous transmission on the uplink. Each of these user terminals transmits its set of processed symbol streams on the uplink to the access point.

(20) At access point **110**, $N_{\text{sub.ap}}$ antennas **224_a** through **224_{ap}** receive the uplink signals from all $N_{\text{sub.up}}$ user terminals transmitting on the uplink. For receive diversity, a transceiver front end **222** may select signals received from one of the antennas **224** for processing. The signals received from multiple antennas **224** may be combined for enhanced receive diversity. The access point's transceiver front end **222** also performs processing complementary to that performed by the user terminal's transceiver front end **254** and provides a recovered uplink data symbol stream. The recovered uplink data symbol stream is an estimate of a data symbol stream $\{s_{\text{sub.up}}\}$ transmitted by a user terminal. An RX data processor **242** processes (e.g., demodulates, deinterleaves, and

decodes) the recovered uplink data symbol stream in accordance with the rate used for that stream to obtain decoded data. The decoded data for each user terminal may be provided to a data sink **244** for storage and/or a controller **230** for further processing.

(21) On the downlink, at access point **110**, a TX data processor **210** receives traffic data from a data source **208** for N.sub.dn user terminals scheduled for downlink transmission, control data from a controller **230** and possibly other data from a scheduler **234**. The various types of data may be sent on different transport channels. TX data processor **210** processes (e.g., encodes, interleaves, and modulates) the traffic data for each user terminal based on the rate selected for that user terminal. TX data processor **210** may provide a downlink data symbol streams for one or more of the N.sub.dn user terminals to be transmitted from one of the N.sub.ap antennas. The transceiver front end **222** receives and processes (e.g., converts to analog, amplifies, filters, and frequency upconverts) the symbol stream to generate a downlink signal. The transceiver front end **222** may also route the downlink signal to one or more of the N.sub.ap antennas **224** for transmit diversity via an RF switch, for example. The controller **230** may control the routing within the transceiver front end **222**. Memory **232** may store data and program codes for the access point **110** and may interface with the controller **230**.

(22) At each user terminal **120**, N.sub.ut,m antennas **252** receive the downlink signals from access point **110**. For receive diversity at the user terminal **120**, the transceiver front end **254** may select signals received from one or more of the antennas **252** for processing. The signals received from multiple antennas **252** may be combined for enhanced receive diversity. The user terminal's transceiver front end **254** also performs processing complementary to that performed by the access point's transceiver front end **222** and provides a recovered downlink data symbol stream. An RX data processor **270** processes (e.g., demodulates, deinterleaves, and decodes) the recovered downlink data symbol stream to obtain decoded data for the user terminal.

(23) In some aspects, the transceiver front end **254** or **222** may include an amplifier and an attenuator having one or more variable resistive elements, as described in more detail herein.

(24) FIG. 3 is a block diagram of an example transceiver front end **300**, such as transceiver front ends **222**, **254** in FIG. 2, in which aspects of the present disclosure may be practiced. The transceiver front end **300** includes at least one transmit (TX) path **302** (also known as a “transmit chain”) for transmitting signals via one or more antennas and at least one receive (RX) path **304** (also known as a “receive chain”) for receiving signals via the antennas. When the TX path **302** and the RX path **304** share an antenna **303**, the paths may be connected with the antenna via an interface **306**, which may include any of various suitable radio frequency (RF) devices, such as a switch, a duplexer, a diplexer, a multiplexer, and the like.

(25) Receiving in-phase (I) and/or quadrature (Q) baseband analog signals from a digital-to-analog converter (DAC) **308**, the TX path **302** may include a baseband filter (BBF) **310**, a mixer **312**, a driver amplifier (DA) **314**, and a power amplifier (PA) **316**. The BBF **310**, the mixer **312**, the DA **314**, and the PA **316** may be included in a radio frequency integrated circuit (RFIC). In some cases, the PA **316** may be external to the RFIC. In some aspects, the input of the DA **314** may be coupled to an attenuator having variable resistive elements, as described in more detail herein. Alternatively or additionally, the attenuator may be coupled to an input of the PA **316**. The attenuator may share the same semiconductor die (e.g., a gallium arsenide (GaAs) die) as the DA **314** and/or the PA **316**.

(26) The BBF **310** filters the baseband signals received from the DAC **308**, and the mixer **312** mixes the filtered baseband signals with a transmit local oscillator (LO) signal to convert the baseband signal of interest to a different frequency (e.g., upconvert from baseband to RF). This frequency-conversion process produces the sum and difference frequencies between the LO frequency and the frequencies of the baseband signal of interest. The sum and difference frequencies are referred to as the “beat frequencies.” The beat frequencies are typically in the RF range, such that the signals output by the mixer **312** are typically RF signals, which may be amplified by the DA **314** and/or by the PA **316** before transmission by the antenna **303**. While one

mixer **312** is illustrated, several mixers may be used to upconvert the filtered baseband signals to one or more intermediate frequencies and to thereafter upconvert the intermediate frequency (IF) signals to a frequency for transmission.

(27) The RX path **304** includes a low noise amplifier (LNA) **322**, a mixer **324**, and a baseband filter (BBF) **326**. The LNA **322**, the mixer **324**, and the BBF **326** may be included in a radio frequency integrated circuit (RFIC), which may or may not be the same RFIC that includes the TX path components. RF signals received via the antenna **303** may be amplified by the LNA **322**, and the mixer **324** mixes the amplified RF signals with a receive local oscillator (LO) signal to convert the RF signal of interest to a different baseband frequency (i.e., downconvert). The baseband signals output by the mixer **324** may be filtered by the BBF **326** before being converted by an analog-to-digital converter (ADC) **328** to digital I and/or Q signals for digital signal processing.

(28) Certain transceivers may employ frequency synthesizers with a variable-frequency oscillator (e.g., a voltage-controlled oscillator (VCO) or a digitally controlled oscillator (DCO)) to generate a stable, tunable LO with a particular tuning range. Thus, the transmit LO frequency may be produced by a TX frequency synthesizer **318**, which may be buffered or amplified by amplifier **320** before being mixed with the baseband signals in the mixer **312**. Similarly, the receive LO frequency may be produced by an RX frequency synthesizer **330**, which may be buffered or amplified by amplifier **332** before being mixed with the RF signals in the mixer **324**. In some cases, a single frequency synthesizer may be used for both the TX path **302** and the RX path **304**.

(29) Example Attenuation Circuit

(30) Some wireless devices may use a multi-mode-multi-band (MMMB) power amplifier (PA), such that a single PA can support multiple modes and multiple frequency bands, to save cost and area. A typical MMMB PA performs well in a high-power mode (HPM) when a signal transmission occurs with high power because the PA may be specifically tuned for HPM. However, in some conditions, the PA may not transmit with high power, and a low-power transmission may be sufficient for a user equipment (UE) to transmit a signal (e.g., to a base station). For a typical PA designed with a 30 dB gain (e.g., implemented via a two-stage or three-stage PA) in HPM, in low-power mode (LPM), the PA gain may be high (e.g., even with a lowered rail voltage (V_{cc}) and reference current (I_{ref}) for the PA), which may cause a problem for meeting a high dynamic range transceiver specification. Implementing an attenuator that allows for further reduction of gain in LPM may reduce HPM performance.

(31) Certain aspects of the present disclosure are directed towards an attenuator (e.g., a gallium arsenide (GaAs) radio-frequency (RF) attenuator) implemented using variable resistive elements. The attenuator allows for maintaining high-power mode (HPM) performance, while allowing for a programmable low-power mode (LPM) gain and maintaining signal linearity. A tunable input attenuator may be used to reduce the overall power gain in the LPM to maintain a signal-to-noise ratio (SNR) at a transmitter output. The attenuator may be turned on in the LPM (e.g., providing over 10 dB attenuation), with little to no impact on input matching and linearity. The attenuator may be turned off in the HPM with little to no insertion loss or effect on input matching and linearity.

(32) FIG. 4 illustrates an example signal processing circuit **400**, in accordance with certain aspects of the present disclosure. The circuit may include an example attenuator **402** and an example impedance transformation circuit **404** coupled between node **450** and node **452**. As shown, the attenuator includes a variable resistive element **406** (e.g., having a first resistance $R_{sub.attn1}$) and a variable resistive element **408** (e.g., having a second resistance $R_{sub.attn2}$), such that the attenuator is programmable. While both resistive elements **406**, **408** are variable resistive elements in the example signal processing circuit **400**, only one of the resistive elements **406**, **408** may be variable in some aspects (not shown). The impedance transformation circuit **404** may be implemented by a quarter-wave impedance transformer used for input impedance matching (e.g., impedance matching by setting an impedance Z_3 looking into the signal processing circuit **400**

from an input path **420**).

(33) The programmable attenuator may be used with little to no effect on input matching. When operating in HPM, the attenuator may be in an off state or otherwise effectively disabled. For example, the variable resistive element **406** and the variable resistive element **408** may be configured with high impedance (e.g., effectively an open circuit). This may be accomplished by implementing each of the variable resistive elements **406**, **408** with one or more transistors, for example.

(34) When operating in LPM, the attenuator may be in an on state (e.g., enabled). For example, the variable resistive element **406** may be biased to provide resistance $R_{\text{sub.attn1}}$. Resistance $R_{\text{sub.attn1}}$ lowers the impedance $Z_{\text{sub.1}}$ looking into an amplification path **410** (e.g., of the DA or the PA), as shown. Due to the impedance transformation circuit **404**, the lower impedance $Z_{\text{sub.1}}$ increases impedance $Z_{\text{sub.2}}$ looking into the impedance transformation circuit **404**. Without compensating for the increase of impedance $Z_{\text{sub.2}}$, impedance $Z_{\text{sub.3}}$ would be increased (e.g., adversely impacting input matching). In some aspects, to compensate, at least adjust, for the increase of impedance $Z_{\text{sub.2}}$, the resistive element **408** may be biased to provide a resistance $R_{\text{sub.attn2}}$ that counteracts the effect of resistance R_{attn1} on impedance $Z_{\text{sub.3}}$. Thus, resistance $R_{\text{sub.attn2}}$ configures impedance $Z_{\text{sub.3}}$ in an impedance-matched condition, while both resistances $R_{\text{sub.attn1}}$ and $R_{\text{sub.attn2}}$ contribute to the attenuation of input signaling (e.g., for the PA).

(35) In some aspects, the tunable attenuator **402** may be integrated with the PA on the same semiconductor die fabricated with a GaAs or indium gallium phosphide (InGaP) heterojunction bipolar transistor (HBT) process, which may provide a more compact and lower cost implementation as compared to conventional complementary metal-oxide-semiconductor (CMOS) implementations (e.g., used in sub-6 GHz RF front-end modules to achieve gain reduction in LPM and reduce SNR degradation at low transmitter output power). When the attenuator **402** is in an off state in HPM, the attenuator **402** may have little to no impact on HPM performance. During the on state in LPM, the attenuator **402** provides attenuation (e.g., greater than 10 dB), which may be independent of input power.

(36) FIG. 5 illustrates example impedance transformation circuits **502**, **506**, in accordance with certain aspects of the present disclosure. As shown, the impedance transformation circuit **404** of FIG. 4 may be implemented using one of the impedance transformation circuits **502**, **506**. Other circuits may be used to implement the impedance transformation circuit **404**.

(37) Implemented as a II-network, the impedance transformation circuit **502** may include: (i) a capacitive element **512** coupled between a first terminal of an inductive element **510** and a reference potential node (e.g., electrical ground) and (ii) a capacitive element **514** coupled between a second terminal of the inductive element **510** and the reference potential node. The first terminal of the inductive element **510** is coupled to the node **450**, and the second terminal of the inductive element **510** is coupled to the node **452**.

(38) Also implemented as a II-network, the impedance transformation circuit **506** may include: (i) an inductive element **522** coupled between a first terminal of a capacitive element **524** and a reference potential node (e.g., electrical ground) and (ii) an inductive element **526** coupled between a second terminal of the capacitive element **524** and the reference potential node. The first terminal of the capacitive element **524** is coupled to the node **450**, and the second terminal of the capacitive element **524** is coupled to the node **452**.

(39) FIG. 6 illustrates an example signal processing circuit **600** including the signal processing circuit **400**, in accordance with certain aspects of the present disclosure. The signal processing circuit **600** may include an amplification path **690** that may receive a bias signal. For example, a transistor **634** (e.g., also referred to herein as a “bias transistor”) may have an emitter coupled to a base of a transistor **660** (e.g., for implementing a DA) through a resistive element **644**. The collector of transistor **634** may be coupled to a voltage rail (e.g., coupled to a power source such as

a battery providing a battery voltage (V_{batt}). A current source **632** may source a reference current (I_{ref}) across resistive element **630**, diodes **638**, **640**, and resistive element **642**, generating a voltage at the base of transistor **634**. Thus, the transistor **634** may be biased, providing a bias signal (e.g., bias current) for biasing the transistor **660** for the DA.

(40) As shown, the amplification path **690** may include a ballasting capacitive element **646**. The collector of transistor **660** may be coupled to a DA output node **698**. The DA output node **698** may be coupled to a base of a transistor **662** (e.g., for implementing a PA) through an inter-stage matching circuit **656** for impedance matching. The collector of transistor **662** may be coupled to an output node **697** (e.g., providing an output RF signal labeled “Rfout”), providing a signal for transmission (e.g., through an output impedance matching circuit **699**). As shown, a reference current may be provided for biasing the PA (e.g., transistor **662**). For example, a transistor **672** may have an emitter coupled to a base of transistor **662** through a resistive element **658**. The collector of transistor **672** may be coupled to the voltage rail. A current source **687** may source a reference current across resistive element **668** and diodes **664**, **666**, generating a voltage at the base of transistor **672**. Thus, the transistor **672** may be biased, providing a bias signal for biasing the transistor **662** for the PA.

(41) In some aspects of the present disclosure, the amplification path **690** may include the signal processing circuit **400** described with respect to FIG. 4. As shown in FIG. 6, the signal processing circuit **400** may be implemented with the impedance transformation circuit **506**, but may be implemented with any suitable impedance transformation circuit. In this example, the signal processing circuit **400** may include an inductive element **522** coupled between a first terminal of a capacitive element **524** and a reference potential node (e.g., electrical ground) and an inductive element **526** coupled between a second terminal of the capacitive element **524** and the reference potential node. The first terminal of the capacitive element **524** is coupled to the input node **651** (e.g., corresponding to node **450** of FIG. 4), and the second terminal of the capacitive element **524** may be coupled to the capacitive element **646**.

(42) The signal processing circuit **400** may include a resistive element **682** and a transistor **686** (e.g., also referred to herein as an “attenuation transistor”) used to implement a variable resistive element (e.g., corresponding to the variable resistive element **406** of FIG. 4). The signal processing circuit **400** may also include a resistive element **680** and a transistor **684** (e.g., also referred to herein as an “attenuation transistor”) used to implement another variable resistive element (e.g., corresponding to the variable resistive element **408** of FIG. 4). The transistors **684**, **686** may be HBTs. The transistors **684**, **686** (and the resistive elements **680**, **682**) may be on the same semiconductor die (a GaAs or InGaP die) as the transistor **660** and/or the transistor **662**. As shown, the collectors of transistors **684**, **686** may be coupled to a voltage rail, as shown. The emitters of transistors **684**, **686** may be coupled to resistive elements **680**, **682**, respectively. The transistor **686** may be biased to set resistance $R_{sub.attn1}$, and the transistor **684** may be biased to set resistance $R_{sub.attn2}$. For example, the signal processing circuit **400** may include a bias circuit having a transistor **612** (e.g., also referred to herein as a “bias transistor”). The transistor **612** may be biased to control the variable resistive elements implemented via transistors **684**, **686**. For example, transistor **612** may have an emitter coupled to bases of transistors **684**, **686**. The collector of transistor **612** may be coupled to the voltage rail, as shown. While a single bias circuit (e.g., bias transistor **612**) is shown for controlling both transistors **684**, **686**, separate bias circuits may be used to individually bias transistors **684**, **686** in some aspects.

(43) A current source **602** may source a reference current to a current path **691** (e.g., across resistive element **604**, diodes **606**, **608**, and resistive element **610**), generating a voltage at the base of transistor **612**. Thus, the transistor **612** may be biased, providing a bias signal for biasing the transistors **684**, **686** to set an attenuation level of the attenuator (e.g., by adjusting the collector-to-emitter impedance of transistors **684**, **686**).

(44) The signal processing circuit **400** may provide a quarter-wave response input matching. Thus,

variable resistive elements **406**, **408** may move the forward transmission (e.g., scattering parameter **S11**) in opposite directions (e.g., on a Smith chart). Thus, the input return loss (IRL) of the signal processing circuit **600** may be maintained with attenuation using the signal processing circuit **400**. (45) FIG. 7 is a flow diagram depicting example operations **700** for signal processing, in accordance with certain aspects of the present disclosure. For example, the operations **700** may be performed by an electrical device such as an attenuation circuit (e.g., attenuator **402**), and in some aspects, a signal processing circuit such as the signal processing circuit **600**.

(46) The operations **700** begin, at block **702**, with the electrical device receiving an input signal (e.g., at input node **651**). At block **704**, the electrical device generates, via an attenuation circuit (e.g., attenuator **402**), an attenuator signal based on the input signal. In some cases (e.g., LPM), the attenuator signal may be an attenuated version of the input signal (e.g., an attenuated signal). In some aspects, the attenuation circuit includes a first variable resistive element (e.g., variable resistive element **408**) coupled between a first node (e.g., node **450** or input node **651**) and a reference potential node. The attenuation circuit may also include a second variable resistive element (e.g., variable resistive element **406**) coupled between a second node (e.g., node **452**) and the reference potential node. An impedance transformation circuit (e.g., impedance transformation circuit **404**) is coupled between the first node and the second node. In some aspects, the electrical device may generate an amplified signal based on the attenuator signal via an amplification transistor (e.g., transistor **660**) coupled to the second node.

(47) In some aspects, the electrical device determines a power mode (e.g., LPM or HPM) associated with signal amplification. The electrical device may control the first variable resistive element and the second variable resistive element based on the power mode. For example, the first variable resistive element may include a first attenuation transistor (e.g., transistor **684**) coupled between the first node and a voltage rail, and the second variable resistive element may include a second attenuation transistor (e.g., transistor **686**) coupled between the second node and the voltage rail. The electrical device may turn off the first attenuation transistor and the second attenuation transistor based on the power mode being a high-power mode (HPM). The electrical device may bias, via a bias circuit, the first attenuation transistor and the second attenuation transistor based on the power mode being a low-power mode (LPM). For instance, the bias circuit may include a bias transistor (e.g., transistor **612**) coupled between the voltage rail and at least one of a base of the first attenuation transistor or a base of the second attenuation transistor. The bias transistor may be coupled between the voltage rail and the bases of the first attenuation transistor and the second attenuation transistor, in some aspects. The electrical device may source, via a current source (e.g., current source **602**) of the bias circuit, a current to a current path (e.g., current path **691**) coupled to a base of the bias transistor.

(48) In some aspects, the first variable resistive element may also include a first resistive element (e.g., resistive element **680**) coupled between the first node and an emitter of the first attenuation transistor. The second variable resistive element may also include a second resistive element (e.g., resistive element **682**) coupled between the second node and an emitter of the second attenuation transistor.

(49) In some aspects, the impedance transformation circuit may include: a first inductive element (e.g., inductive element **522**) coupled between the first node and the reference potential node, a second inductive element (e.g., inductive element **526**) coupled between the second node and the reference potential node, and a capacitive element (e.g., capacitive element **524**) coupled between the first node and the second node.

(50) In some aspects, the impedance transformation circuit may include: a first capacitive element (e.g., capacitive element **512**) coupled between the first node and the reference potential node, a second capacitive element (e.g., capacitive element **514**) coupled between the second node and the reference potential node, and an inductive element (e.g., inductive element **510**) coupled between the first node and the second node.

Example Aspects

- (51) Aspect 1: A signal processing circuit, comprising: an impedance transformation circuit coupled between a first node and a second node; and an attenuation circuit including: a first variable resistive element coupled between the first node and a reference potential node; and a second variable resistive element coupled between the second node and the reference potential node.
- (52) Aspect 2: The signal processing circuit of Aspect 1, wherein the impedance transformation circuit includes: a first inductive element coupled between the first node and the reference potential node; a second inductive element coupled between the second node and the reference potential node; and a capacitive element coupled between the first node and the second node.
- (53) Aspect 3: The signal processing circuit of Aspect 1 or 2, wherein the impedance transformation circuit includes: a first capacitive element coupled between the first node and the reference potential node; a second capacitive element coupled between the second node and the reference potential node; and an inductive element coupled between the first node and the second node.
- (54) Aspect 4: The signal processing circuit according to any of Aspects 1-3, wherein: the first variable resistive element includes a first attenuation transistor coupled between the first node and a voltage rail; and the second variable resistive element includes a second attenuation transistor coupled between the second node and the voltage rail.
- (55) Aspect 5: The signal processing circuit of Aspect 4, further comprising a bias transistor coupled between the voltage rail and at least one of a base of the first attenuation transistor or a base of the second attenuation transistor.
- (56) Aspect 6: The signal processing circuit of Aspect 5, wherein the bias transistor is coupled between the voltage rail and the bases of the first attenuation transistor and the second attenuation transistor.
- (57) Aspect 7: The signal processing circuit of Aspect 5 or 6, further comprising a current source coupled to a current path, the current path being coupled to a base of the bias transistor.
- (58) Aspect 8: The signal processing circuit according to any of Aspects 4-7, wherein: the first variable resistive element further comprises a first resistive element coupled between the first node and an emitter of the first attenuation transistor; and the second variable resistive element further comprises a second resistive element coupled between the second node and an emitter of the second attenuation transistor.
- (59) Aspect 9: The signal processing circuit according to any of Aspects 1-8, further comprising: an amplification transistor; and a capacitive element coupled between the attenuation circuit and a base of the amplification transistor.
- (60) Aspect 10: The signal processing circuit according to any of Aspects 1-9, wherein: in a high-power mode (HPM) associated with the signal processing circuit, at least one of the first variable resistive element or the second variable resistive element is configured to have a first resistance; and in a low-power mode (LPM) associated with the signal processing circuit, at least one of the first variable resistive element or the second variable resistive element is configured to have a second resistance less than the first resistance.
- (61) Aspect 11: A wireless device, comprising: a controller configured to determine a power mode associated with signal transmission; an attenuator comprising: a first variable resistive element coupled between a first node and a reference potential node; and a second variable resistive element coupled between a second node and the reference potential node, wherein the controller is configured to bias the first variable resistive element and the second variable resistive element to generate an attenuator signal based on an input signal; an impedance transformation circuit coupled between the first node and the second node; and an amplifier coupled to the attenuator and configured to generate a signal for transmission based on the attenuator signal.
- (62) Aspect 12: The wireless device of Aspect 11, wherein: the first variable resistive element includes a first attenuation transistor coupled between the first node and a voltage rail; and the

second variable resistive element includes a second attenuation transistor coupled between the second node and the voltage rail.

(63) Aspect 13: The wireless device of Aspect 12, wherein the controller is further configured to turn off the first attenuation transistor and the second attenuation transistor based on the power mode being a high-power mode (HPM).

(64) Aspect 14: The wireless device of Aspect 12 or 13, wherein, to bias the first variable resistive element and the second variable resistive element, the controller is configured to bias, via a bias circuit, the first attenuation transistor and the second attenuation transistor based on the power mode being a low-power mode (LPM).

(65) Aspect 15: The wireless device of Aspect 14, wherein the bias circuit comprises a bias transistor coupled between a voltage rail and at least one of a base of the first attenuation transistor or a base of the second attenuation transistor.

(66) Aspect 16: The wireless device of Aspect 15, wherein the bias transistor is coupled between the voltage rail and the bases of the first attenuation transistor and the second attenuation transistor.

(67) Aspect 17: The wireless device of Aspect 15 or 16, wherein, to bias the first variable resistive element and the second variable resistive element, the controller is configured to control a current source to provide a current to a current path coupled to a base of the bias transistor.

(68) Aspect 18: A method for signal processing, comprising: receiving an input signal; and generating, via an attenuation circuit, an attenuator signal based on the input signal, the attenuation circuit comprising: a first variable resistive element coupled between a first node and a reference potential node; and a second variable resistive element coupled between a second node and the reference potential node, wherein an impedance transformation circuit is coupled between the first node and the second node.

(69) Aspect 19: The method of Aspect 18, further comprising generating an amplified signal based on the attenuator signal via an amplification transistor coupled to the second node.

(70) Aspect 20: The method of Aspect 18 or 19, further comprising: determining a power mode associated with signal amplification; and controlling the first variable resistive element and the second variable resistive element based on the power mode.

(71) Aspect 21: The method of Aspect 20, wherein: the first variable resistive element includes a first attenuation transistor coupled between the first node and a voltage rail; and the second variable resistive element includes a second attenuation transistor coupled between the second node and the voltage rail.

(72) Aspect 22: The method of Aspect 21, further comprising turning off the first attenuation transistor and the second attenuation transistor based on the power mode being a high-power mode (HPM).

(73) Aspect 23: The method of Aspect 21 or 22, further comprising biasing, via a bias circuit, the first attenuation transistor and the second attenuation transistor based on the power mode being a low-power mode (LPM).

(74) Aspect 24: The method of Aspect 23, wherein the bias circuit comprises a bias transistor coupled between the voltage rail and at least one of a base of the first attenuation transistor or a base of the second attenuation transistor.

(75) Aspect 25: The method of Aspect 24, wherein the bias transistor is coupled between the voltage rail and the bases of the first attenuation transistor and the second attenuation transistor.

(76) Aspect 26: The method of Aspect 24 or 25, wherein biasing the first attenuation transistor and the second attenuation transistor includes sourcing, via a current source of the bias circuit, a current to a current path coupled to a base of the bias transistor.

(77) Aspect 27: The method according to any of Aspects 21-26, wherein: the first variable resistive element further comprises a first resistive element coupled between the first node and an emitter of the first attenuation transistor; and the second variable resistive element further comprises a second resistive element coupled between the second node and an emitter of the second attenuation

transistor.

(78) Aspect 28: The method according to any of Aspects 18-27, wherein the impedance transformation circuit includes: a first inductive element coupled between the first node and the reference potential node; a second inductive element coupled between the second node and the reference potential node; and a capacitive element coupled between the first node and the second node.

(79) Aspect 29: The method according to any of Aspects 18-28, wherein the impedance transformation circuit includes: a first capacitive element coupled between the first node and the reference potential node; a second capacitive element coupled between the second node and the reference potential node; and an inductive element coupled between the first node and the second node.

(80) Within the present disclosure, the word “exemplary” is used to mean “serving as an example, instance, or illustration.” Any implementation or aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects of the disclosure. Likewise, the term “aspects” does not require that all aspects of the disclosure include the discussed feature, advantage, or mode of operation. The term “coupled” is used herein to refer to the direct or indirect coupling between two objects. For example, if object A physically touches object B and object B touches object C, then objects A and C may still be considered coupled to one another—even if objects A and C do not directly physically touch each other. For instance, a first object may be coupled to a second object even though the first object is never directly physically in contact with the second object. The terms “circuit” and “circuitry” are used broadly and intended to include both hardware implementations of electrical devices and conductors that, when connected and configured, enable the performance of the functions described in the present disclosure, without limitation as to the type of electronic circuits.

(81) The apparatus and methods described in the detailed description are illustrated in the accompanying drawings by various blocks, modules, components, circuits, steps, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using hardware, for example.

(82) One or more of the components, steps, features, and/or functions illustrated herein may be rearranged and/or combined into a single component, step, feature, or function or embodied in several components, steps, or functions. Additional elements, components, steps, and/or functions may also be added without departing from features disclosed herein. The apparatus, devices, and/or components illustrated herein may be configured to perform one or more of the methods, features, or steps described herein.

(83) It is to be understood that the specific order or hierarchy of steps in the methods disclosed is an illustration of exemplary processes. Based upon design preferences, it is understood that the specific order or hierarchy of steps in the methods may be rearranged. The accompanying method claims present elements of the various steps in a sample order, and are not meant to be limited to the specific order or hierarchy presented unless specifically recited therein.

(84) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language of the claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. A phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover at least: a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c). All

structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112 (f) unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.” For example, means for determining may a digital controller such as digital controller **510**. (85) It is to be understood that the claims are not limited to the precise configuration and components illustrated above. Various modifications, changes and variations may be made in the arrangement, operation and details of the methods and apparatus described above without departing from the scope of the claims.

Claims

1. A signal processing circuit, comprising: an impedance transformation circuit coupled between a first node and a second node; and an attenuation circuit including: a first variable resistive element including a first attenuation transistor coupled between the first node and a voltage rail; and a second variable resistive element including a second attenuation transistor coupled between the second node and the voltage rail.
2. The signal processing circuit of claim 1, wherein the impedance transformation circuit includes: a first inductive element coupled between the first node and a reference potential node; a second inductive element coupled between the second node and the reference potential node; and a capacitive element coupled between the first node and the second node.
3. The signal processing circuit of claim 1, wherein the impedance transformation circuit includes: a first capacitive element coupled between the first node and a reference potential node; a second capacitive element coupled between the second node and the reference potential node; and an inductive element coupled between the first node and the second node.
4. The signal processing circuit of claim 1, further comprising a bias transistor coupled between the voltage rail and at least one of a base of the first attenuation transistor or a base of the second attenuation transistor.
5. The signal processing circuit of claim 4, wherein the bias transistor is coupled between the voltage rail and the bases of the first attenuation transistor and the second attenuation transistor.
6. The signal processing circuit of claim 4, further comprising a current source coupled to a current path, the current path being coupled to a base of the bias transistor.
7. The signal processing circuit of claim 1, wherein: the first variable resistive element further comprises a first resistive element coupled between the first node and an emitter of the first attenuation transistor; and the second variable resistive element further comprises a second resistive element coupled between the second node and an emitter of the second attenuation transistor.
8. The signal processing circuit of claim 1, further comprising: an amplification transistor; and a capacitive element coupled between the attenuation circuit and a base of the amplification transistor.
9. The signal processing circuit of claim 1, wherein: in a high-power mode (HPM) associated with the signal processing circuit, at least one of the first variable resistive element or the second variable resistive element is configured to have a first resistance; and in a low-power mode (LPM) associated with the signal processing circuit, at least one of the first variable resistive element or the second variable resistive element is configured to have a second resistance less than the first resistance.
10. A wireless device, comprising: a controller configured to determine a power mode associated with signal transmission; an attenuator comprising: a first variable resistive element coupled

between a first node and a reference potential node or a voltage rail; and a second variable resistive element coupled between a second node and the reference potential node or the voltage rail, wherein the controller is configured to bias the first variable resistive element and the second variable resistive element to generate an attenuator signal based on an input signal; an impedance transformation circuit coupled between the first node and the second node; and an amplifier coupled to the attenuator and configured to generate a signal for transmission based on the attenuator signal.

11. The wireless device of claim 10, wherein: the first variable resistive element includes a first attenuation transistor coupled between the first node and the voltage rail; and the second variable resistive element includes a second attenuation transistor coupled between the second node and the voltage rail.

12. The wireless device of claim 11, wherein the controller is further configured to turn off the first attenuation transistor and the second attenuation transistor based on the power mode being a high-power mode (HPM).

13. The wireless device of claim 11, wherein, to bias the first variable resistive element and the second variable resistive element, the controller is configured to bias, via a bias circuit, the first attenuation transistor and the second attenuation transistor based on the power mode being a low-power mode (LPM).

14. The wireless device of claim 13, wherein the bias circuit comprises a bias transistor coupled between the voltage rail and at least one of a base of the first attenuation transistor or a base of the second attenuation transistor.

15. The wireless device of claim 14, wherein the bias transistor is coupled between the voltage rail and the bases of the first attenuation transistor and the second attenuation transistor.

16. The wireless device of claim 14, wherein, to bias the first variable resistive element and the second variable resistive element, the controller is configured to control a current source to provide a current to a current path coupled to a base of the bias transistor.

17. A method for signal processing, comprising: receiving an input signal; and generating, via an attenuation circuit, an attenuator signal based on the input signal, the attenuation circuit comprising: a first variable resistive element including a first attenuation transistor coupled between a first node and a voltage rail; and a second variable resistive element including a second attenuation transistor coupled between a second node and the voltage rail, wherein an impedance transformation circuit is coupled between the first node and the second node.

18. The method of claim 17, further comprising generating an amplified signal based on the attenuator signal via an amplification transistor coupled to the second node.

19. The method of claim 17, further comprising: determining a power mode associated with signal amplification; and controlling the first variable resistive element and the second variable resistive element based on the power mode.

20. The method of claim 19, further comprising turning off the first attenuation transistor and the second attenuation transistor based on the power mode being a high-power mode (HPM).

21. The method of claim 19, further comprising biasing, via a bias circuit, the first attenuation transistor and the second attenuation transistor based on the power mode being a low-power mode (LPM).

22. The method of claim 21, wherein the bias circuit comprises a bias transistor coupled between the voltage rail and at least one of a base of the first attenuation transistor or a base of the second attenuation transistor.

23. The method of claim 22, wherein the bias transistor is coupled between the voltage rail and the bases of the first attenuation transistor and the second attenuation transistor.

24. The method of claim 22, wherein biasing the first attenuation transistor and the second attenuation transistor includes sourcing, via a current source of the bias circuit, a current to a current path coupled to a base of the bias transistor.

25. The method of claim 17, wherein: the first variable resistive element further comprises a first resistive element coupled between the first node and an emitter of the first attenuation transistor; and the second variable resistive element further comprises a second resistive element coupled between the second node and an emitter of the second attenuation transistor.

26. The method of claim 17, wherein the impedance transformation circuit includes: a first inductive element coupled between the first node and a reference potential node; a second inductive element coupled between the second node and the reference potential node; and a capacitive element coupled between the first node and the second node.

27. The method of claim 17, wherein the impedance transformation circuit includes: a first capacitive element coupled between the first node and a reference potential node; a second capacitive element coupled between the second node and the reference potential node; and an inductive element coupled between the first node and the second node.
